



Technical Service Bulletin

Technical Service Bulletin: TSB100978

Released Date: 02-Jun-2021

New Modular Common Rail System (MCRS) Fuel System

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

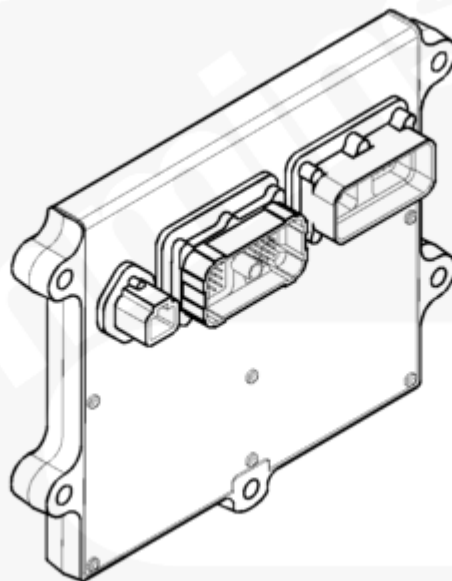
Contents

This document announces a new MCRS fuel system that includes new injectors, fuel pump, wiring harnesses, engine control module (ECM), calibration, and service information. This new system is commonly referred to as a C3 fuel system. It provides improved robustness. This fuel system will be initially released on Industrial applications, Marine applications, and the QSK50 CM2150 Power Generation engines with Advanced Engine Monitoring system.

Note : Updating C2 style injectors to the C3 style injectors is not a requirement.

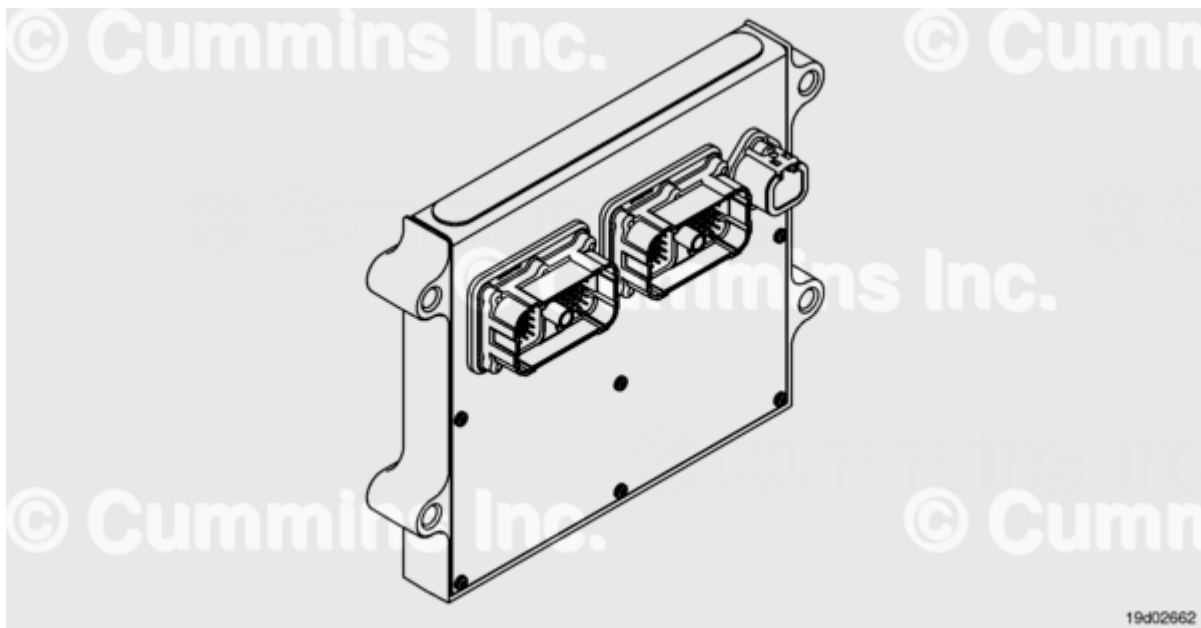
When servicing an engine, the ESN **must** be checked to determine the fuel system type. The fuel system type can be determined by visually checking the ECM.

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Old Style - CM850 (Part Number 4921797)



New Style - CM2150 (Part Number 4995445)

New style fuel system (C3) parts can **not** be interchanged with previous fuel system parts. Production engines will be transitioning from the old style to the new style fuel system over a period of time. The old style fuel system parts **must** remain in service for existing engines. See Table 1 for a list of new fuel system part numbers. Included with the Marine changes is an available customer option for a thermostat insert for keel cooled Marine QSK50 CM2150 engines, to reduce the external flow from the engine to the keel coolers.

The service procedures have been modified for all MCRS fuel systems to include steps to relieve the fuel pressure when servicing the fuel system.

Note : Extension harness, Part Number 2881173, must be used when installing a fast IMV pump on a slow IMV engine to bridge the gap in the wiring harness.

- Use the following procedure in Service Manual, QSK19, QSK19 CM850 MCRS, QSK19 CM2150 MCRS, Bulletin 4021592. Refer to Procedure 006-051 in Section 6. (</qs3/pubsys2/xml/en/procedures/20/20-006-051-tr.html>)
- Use the following procedure in Service Manual, K38, K50, QSK38, and QSK50, Bulletin 4021528. Refer to Procedure 006-051 in Section 6. (</qs3/pubsys2/xml/en/procedures/28/28-006-051-tr.html>)
- Use the following procedure in Service Manual, QSK45 and QSK60, Bulletin 4021530. Refer to Procedure 006-051 in Section 6 (</qs3/pubsys2/xml/en/procedures/56/56-006-051-tr.html>)
- Use the following procedure in the Troubleshooting and Repair Manual, QSK38, QSK50, and QSK60 CM850 Modular Common Rail System, Bulletin 4021533. Refer to Procedure 019-115 in Section 19. (</qs3/pubsys2/xml/en/procedures/102/102-019-115.html>)
- Use the following procedure in the Troubleshooting and Repair Manual, QSK38, QSK50, and QSK60 CM2150 Modular Common Rail System, Bulletin 4022102. Refer to Procedure 019-115 in Section 19. (</qs3/pubsys2/xml/en/procedures/102/102-019-115.html>)
- Use the following procedure in the Troubleshooting and Repair Manual, Electronic Control System QSK19 CM850 Modular Common Rail System, Bulletin 4021493. Refer to Procedure 019-115 in Section 19. (</qs3/pubsys2/xml/en/procedures/102/102-019-115.html>)

- Use the following procedure in the Troubleshooting and Repair Manual, QSK19 CM2150 Electronic Control System, Bulletin 4022094. Refer to Procedure 019-115 in Section 19. (</qs3/pubsys2/xml/en/procedures/102/102-019-115.html>)

The new engine wiring harnesses require new wiring diagrams and Troubleshooting and Repair manuals with updated circuit art. Troubleshooting procedures are common for both fuel systems. See Table 2 for a list of new publication bulletin numbers.

When servicing engines with the new style fuel system, INSITE™ electronic service tool version 7.4 is required. There is also an ECM benchtop harness specific for the CM2150 ECM (Service Part Number 4918583).

Training beyond the existing MCRS training is **not** required.

Table 1

Part Description	Part Name	Part Number
QSK19 Tier 3 Injector	INJECTOR (Quantity: 6)	2867149
QSK19 Tier 2/QSK38/50 Injector	INJECTOR (Quantity: QSK19: 6/QSK38:12/QSK50:16)	2867147
QSK60 Injector	INJECTOR (Quantity: 16)	2867148
CM2150 ECM	MODULE, ELECTRONIC CONTROL	4995445
CM2150 ECM Mounting Bolts	SCREW, HEX FLANGE HEAD CAP	3900633
QSK50/60 CM2150 ECM Bracket	SUPPORT, ETR CNT MODULE	2867198
QSK60 Breather	HOSE, MOLDED	2871205
QSK38/50/60 Electronic Fuel Control Actuator*	ACTUATOR, ETR FUEL CONTROL	2882025
QSK38/50/60 Electronic Fuel Control Actuator Main Wiring Harness*	HARNESS, WRG	2881173
QSK38/50/60 Electronic Fuel Control Actuator Main Wiring Harness*	HARNESS, WRG	2881173
QSK38 Fuel Pump*	PUMP, FUEL INJECTION	2881171
QSK50/60 Fuel Pump*	PUMP, FUEL INJECTION	2881172
QSK19 Industrial Standard Fuel Filter Main Wiring Harness	HARNESS, ETR CNT MDL WRG	4969189
QSK19 Industrial Remote Fuel Filter Main Wiring Harness	HARNESS, ETR CNT MDL WRG	2873778

Table 1		
Part Description	Part Name	Part Number
QSK19 Marine Classed Main Wiring Harness	HARNESS, ETR CNT MDL WRG	2873266
QSK19 Marine Non-Classed Main Wiring Harness	HARNESS, ETR CNT MDL WRG	2867065
QSK38 Industrial Left Bank Wiring Harness	HARNESS, ETR CNT MDL WRG	3642434
QSK38 Industrial Left Bank Front Main Wiring Harness*	HARNESS, ETR CNT MDL WRG	2881591
QSK38 Industrial Right Bank Harness	HARNESS, ETR CNT MDL WRG	3642408
QSK38 Industrial Left Bank Injector Main Wiring Harness	HARNESS, ETR CNT MDL WRG	3642409
QSK38 Industrial Right Bank Injector Main Wiring Harness	HARNESS, ETR CNT MDL WRG	3642410
QSK38 Marine Left Bank Base Wiring Harness	HARNESS, ETR CNT MDL WRG	2862877
QSK38 Marine Left Bank Premium Wiring Harness	HARNESS, ETR CNT MDL WRG	2862879
QSK38 Marine Left Bank Classed Wiring Harness	HARNESS, ETR CNT MDL WRG	2862881
QSK38 Marine Right Bank Base Wiring Harness	HARNESS, ETR CNT MDL WRG	2862878
QSK38 Marine Right Bank Premium Wiring Harness	HARNESS, ETR CNT MDL WRG	2862880
QSK38 Marine Right Bank Classed Wiring Harness	HARNESS, ETR CNT MDL WRG	2862882
QSK38 Marine Left Bank Rear Injector Main Wiring Harness	HARNESS, ETR CNT MDL WRG	2862875
QSK38 Marine Right Bank Rear Injector Main Wiring Harness and QSK38/50 Fuel Filter Main Wiring Harness	HARNESS, ETR CNT MDL WRG	2862876
QSK38 Marine Left Bank Rear Injector/EGT Main Wiring Harness	HARNESS, ETR CNT MDL WRG	2862873
QSK38 Marine Right Bank Rear Injector/EGT Main Wiring Harness	HARNESS, ETR CNT MDL WRG	2862874

Table 1		
Part Description	Part Name	Part Number
QSK50 Industrial Left Bank Harness	HARNESS, ETR CNT MDL WRG	2873839
QSK50 Industrial Left Bank Harness Compatible with Delco™ Starter in 4th Quadrant (ST6160)	HARNESS, ETR CNT MDL WRG	2881346
QSK50 Industrial Left Bank Front Main Wiring Harness*	HARNESS, ETR CNT MDL WRG	2881589
QSK50 Industrial Right Bank Harness	HARNESS, ETR CNT MDL WRG	3642412
QSK50 Industrial Left Bank Injector / EGT Main Wiring Harness	HARNESS, ETR CNT MDL WRG	3642413
QSK50 Industrial Right Bank Injector / EGT Main Wiring Harness	HARNESS, ETR CNT MDL WRG	3642414
QSK38/50 Industrial Thermistor EGT*	SENSOR, TEMPERATURE	4326918
QSK50 Marine Left Bank Base Wiring Harness	HARNESS, ETR CNT MDL WRG	2862869
QSK50 Marine Left Bank Premium Wiring Harness	HARNESS, ETR CNT MDL WRG	2862868
QSK50 Marine Left Bank Classed Wiring Harness	HARNESS, ETR CNT MDL WRG	2862885
QSK50 Marine Right Bank Base Wiring Harness	HARNESS, ETR CNT MDL WRG	2862884
QSK50 Marine Right Bank Premium Wiring Harness	HARNESS, ETR CNT MDL WRG	2862849
QSK50 Marine Right Bank Classed Wiring Harness	HARNESS, ETR CNT MDL WRG	2862886
QSK38/50 Marine Left Bank Front Main Wiring Harness*	HARNESS, ETR CNT MDL WRG	2881343
QSK38/50 Marine Left Bank Injector Main Wiring Harness	HARNESS, ETR CNT MDL WRG	4998577
QSK38/50 Marine Right Bank Injector Main Wiring Harness	HARNESS, ETR CNT MDL WRG	4998578
QSK38/50 Marine Left Bank Injector / EGT Main Wiring Harness	HARNESS, ETR CNT MDL WRG	4998575

Table 1		
Part Description	Part Name	Part Number
QSK Industrial Thermistor EGT*	SENSOR, TEMPERATURE	4326918
QSK38/50 Marine Right Bank Injector / EGT Main Wiring Harness	HARNESS, ETR CNT MDL WRG	4998576
QSK38/50 Marine Left Bank Front / Front Mount Fuel Filter Main Wiring Harness*	HARNESS, ETR CNT MDL WRG	2881344
QSK38/50 Marine Fuel Filter Main Wiring Harness	HARNESS, ETR CNT MDL WRG	4998522
QSK50 Power Generation with Advanced Engine Monitoring Right Bank Harness	HARNESS, ETR CNT MDL WRG	2880749
QSK50 Power Generation with Advanced Engine Monitoring Left Bank Harness	HARNESS, ETR CNT MDL WRG	2891235
QSK60 Industrial Left Bank Harness	HARNESS, ETR CNT MDL WRG	2873840
QSK60 Industrial Left Bank Front Main Wiring Harness with Remote Fuel Filter*	HARNESS, ETR CNT MDL WRG	2881588
QSK60 Industrial Right Bank Harness	HARNESS, ETR CNT MDL WRG	3642416
QSK60 Industrial Left Bank Injector / EGT Main Wiring Harness	HARNESS, ETR CNT MDL WRG	3642417
QSK60 Industrial Right Bank Injector /EGT Main Wiring Harness	HARNESS, ETR CNT MDL WRG	3642418
QSK60 Industrial Thermistor EGT*	SENSOR, TEMPERATURE	4954450
QSK60 Marine Left Bank Premium Wiring Harness	HARNESS, ETR CNT MDL WRG	2870658
QSK60 Marine Left Bank Classed Wiring Harness	HARNESS, ETR CNT MDL WRG	2870660
QSK60 Marine Right Bank Premium Wiring Harness	HARNESS, ETR CNT MDL WRG	2870659
QSK60 Marine Right Bank Classed Wiring Harness	HARNESS, ETR CNT MDL WRG	2870661

Table 1		
Part Description	Part Name	Part Number
QSK60 Marine Left Bank Front Main Wiring Harness*	HARNESS, ETR CNT MDL WRG	2881345
QSK60 Marine Left Bank Injector Main Wiring Harness	HARNESS, ETR CNT MDL WRG	2870654
QSK60 Marine Right Bank Injector Main Wiring Harness	HARNESS, ETR CNT MDL WRG	2870655
QSK60 Marine Left Bank Injector / EGT Main Wiring Harness	HARNESS, ETR CNT MDL WRG	2870652
QSK60 Marine Right Bank Injector / EGT Main Wiring Harness	HARNESS, ETR CNT MDL WRG	2870653
QSK60 Marine Fuel Filter Main Wiring Harness	HARNESS, ETR CNT MDL WRG	2870611
QSK60 Offshore Drill Rig Air Shutoff Valve Main Wiring Harness	HARNESS, ETR CNT MDL WRG	2870697
QSK60 Marine Stack Temperature Main Wiring Harness	HARNESS, ETR CNT MDL WRG	2870699
Marine Non-Classed OEM 10 Meter Main Wiring Harness	HARNESS, WIRING	2880932
Marine Classed OEM 10 Meter Main Wiring Harness to CIB	HARNESS, WIRING	2880933
Marine Non-Classed OEM 5 Meter Main Wiring Harness to CIB	HARNESS, WIRING	2880934
Marine Classed OEM 5 Meter Main Wiring Harness to CIB	HARNESS, WIRING	2880935
QSK50 Marine Keel Cooled Thermostat Insert	INSERT, THERMOSTAT	2873714
QSK19 Power Generation Main Wiring Harness	HARNESS, ETR CNT MDL WRG	2891042
QSK38 Power Generation Left Bank Main Wiring Harness	HARNESS, ETR CNT MDL WRG	2891081
QSK38 Power Generation Right Bank Main Wiring Harness	HARNESS, ETR CNT MDL WRG	2881187

Table 1		
Part Description	Part Name	Part Number
QSK50 Power Generation Left Bank Main Wiring Harness	HARNESS, ETR CNT MDL WRG	3649349
QSK50 Power Generation Right Bank Main Wiring Harness	HARNESS, ETR CNT MDL WRG	2891146
QSK60 Power Generation Left Bank Main Wiring Harness	HARNESS, ETR CNT MDL WRG	2881188
QSK60 Power Generation Right Bank Main Wiring Harness	HARNESS, ETR CNT MDL WRG	2886094
QSK19/38/50 Power Generation Injector – No Flow Limiter	INJECTOR (Quantity: QSK19: 6/QSK38:12/QSK50:16)	2881088
QSK19/38/50 Power Generation Injector – No Flow Limiter	INJECTOR KIT	2882129 or 2882129NX
QSK60 Power Generation Injector – No Flow Limiter	INJECTOR (Quantity: 16)	2881089
QSK60 Power Generation Injector – No Flow Limiter	INJECTOR KIT	2882130 or 2882130NX

*If up-fitting an engine from CM850 to CM2150 the Fuel Pump and Front Main Wiring Harnesses can be reused. The Electronic Fuel Control Actuator is required to up-fit the fuel pump. The Electronic Fuel Control Actuator Main Wiring Harness may be required to reach the Electronic Fuel Control Acutator. Thermistor EGTs may be required if **not** already installed on the engine.

Table 2	
Publication Title	Bulletin Number
QSK19 CM2150 Electronic Control System Troubleshooting and Repair Manual	4022094
QSK38/50/60 CM2150 Electronic Control System Troubleshooting and Repair Manual	4022102
QSK19 CM2150 Industrial Modular Common Rail System Wiring Diagram	4022096
QSK19 CM2150 Modular Common Rail System Marine Propulsion with C Command™ Panel System Wiring Diagram	4022098

Table 2

Publication Title	Bulletin Number
QSK19 CM2150 Modular Common Rail System Marine Propulsion with C Command Elite™ and C Command Elite Plus™ Panel System Wiring Diagram	4022099
QSK19 CM2150 Modular Common Rail System Marine Auxiliary with C Command™ Panel System Wiring Diagram	4022100
QSK19 CM2150 Modular Common Rail System Marine Auxiliary with C Command Elite™ and C Command Elite Plus™ Panel System Wiring Diagram	4022101
QSK38 CM2150 Industrial Modular Common Rail System Wiring Diagram	4022104
QSK38 CM2150 Modular Common Rail System Marine Propulsion with C Command™ Panel System Wiring Diagram	4022106
QSK38 CM2150 Modular Common Rail System Marine Propulsion with C Command Elite™ and C Command Elite Plus™ Panel System Wiring Diagram	4022107
QSK38 CM2150 Modular Common Rail System Marine Auxiliary with C Command™ Panel System Wiring Diagram	4022108
QSK38 CM2150 Modular Common Rail System Marine Auxiliary with C Command Elite™ and C Command Elite Plus™ Panel System Wiring Diagram	4022109
QSK50 CM2150 Industrial Modular Common Rail System Wiring Diagram	4022110
QSK50 CM2150 Modular Common Rail System Marine Propulsion with C Command™ Panel System Wiring Diagram	4022112
QSK50 CM2150 Modular Common Rail System Marine Propulsion with C Command Elite™ and C Command Elite Plus™ Panel System Wiring Diagram	4022113
QSK50 CM2150 Modular Common Rail System Marine Auxiliary with C Command™ Panel System Wiring Diagram	4022114

Table 2	
Publication Title	Bulletin Number
QSK50 CM2150 Modular Common Rail System Marine Auxiliary with C Command Elite™ and C Command Elite Plus™ Panel System Wiring Diagram	4022115
QSK50 CM2150 Power Generation MCRS (Advanced Engine Monitoring) Wiring Diagram	2883461
QSK60 CM2150 Industrial Modular Common Rail System Wiring Diagram	4022116
QSK60 CM2150 Modular Common Rail System Marine Propulsion with C Command™ Panel System Wiring Diagram	4022118
QSK60 CM2150 Modular Common Rail System Marine Propulsion with C Command Elite™ and C Command Elite Plus™ Panel System Wiring Diagram	4022119
QSK60 CM2150 Modular Common Rail System Marine Auxiliary with C Command™ Panel System Wiring Diagram	4022120
QSK60 CM2150 Modular Common Rail System Marine Auxiliary with C Command Elite™ and C Command Elite Plus™ Panel System Wiring Diagram	4022121

Document History

Date	Details
2010-12-15	Module Created
2010-12-15	Information has been added for newly released QSK50 CM2150 Power Generation with Advanced Engine Monitoring System engines.
2011-3-22	Re-wording of part names to be more clear
2011-6-10	Update part numbers
2011-7-20	Metadata updated
2012-4-17	QSOL Quick Fix Reason: Hyperlink Error Notes: none

Date	Details
2013-6-26	QSOL Quick Fix Reason: Format Error Notes: Extension Harness should be Main Wiring Harness
2013-10-22	Add additional parts required for CM850 to CM2150 Up-fit
2015-3-25	Update part numbers
2019-4-9	Added extension harness part number and note
2021-5-28	Added note. Updated Table 1.

Last Modified: 02-Jun-2021
